

Affine Gap Alignment:

Inputs: $\langle X, Y, \alpha, \mu, \gamma, \tau \rangle$

- X, Y = strings of length m, n with characters indexed by i, j respectively
- α = match score
- μ = mismatch penalty
- γ = gap opening penalty
- τ = gap extension penalty (single-letter gap penalty)

Output: an alignment which maximizes the following score:

$$\alpha(\# \text{ matches}) - \mu(\# \text{ mismatches}) - \gamma(\# \text{ gap clusters}) - \tau(\# \text{ single gaps})$$

Def: Auxiliary Data Structures

Matrices V, G, E, F

- V = best-score matrix
- G = Match-mismatch matrix
- E = X -gap matrix
- F = Y -gap matrix

ex. $X = \text{ATCGGC}$ $Y = \text{AGC}$

- $\alpha = 2$ (score = +2)
- $\mu = 1$ (penalty = -1)
- $\gamma = 2$ (penalty = -2)
- $\tau = 1$ (penalty = -1)

pseudocode: $V(0,0) \leftarrow 0$ for $i \leftarrow 1$ to m do $V(i,0) = E(i,0) \leftarrow -\gamma - i\tau$

end

for $j \leftarrow 1$ to n do $V(0,j) = F(0,j) \leftarrow -\gamma - j\tau$

end

for $i \leftarrow 1$ to $m, j \leftarrow 1$ to n do

$$G(i,j) = \begin{cases} V(i-1,j-1) + \alpha & \text{if } x_i = y_j \\ V(i-1,j-1) - \mu & \text{if } x_i \neq y_j \end{cases}$$

$$E(i,j) \leftarrow \max \begin{cases} E(i,j-1) - \tau \\ V(i,j-1) - \gamma - \tau \end{cases}$$

$$F(i,j) \leftarrow \max \begin{cases} F(i-1,j) - \tau \\ V(i-1,j) - \gamma - \tau \end{cases}$$

$$V(i,j) \leftarrow \max \begin{cases} G(i,j) \\ E(i,j) \\ F(i,j) \end{cases}$$

$$\begin{matrix} \alpha = 2 & \gamma = 2 \\ \mu = 1 & \tau = 1 \end{matrix}$$

Random Biosequences

$$\text{Prob}(A) = \frac{4}{12}$$

$$\text{Prob}(B) = \frac{5}{12}$$

$$\text{Prob}(C) = \frac{3}{12}$$

ex.

A	A	B
B	A	C
B	B	C
C	B	A

How is it that it stops at length $O(\log n)$ for optimal local alignment?Answer: the very special scoring scheme

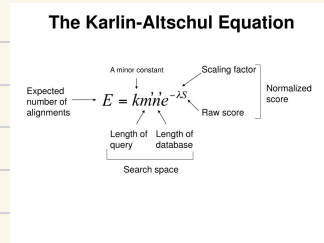
Karlin-Altschul Statistics (1990)

• only for ungapped alignments

5 Fundamental Axioms:

1. Scoring matrix this at length 1 positive score
2. expected score of the scoring matrix needs to be negative
3. the letters of the random sequences are independent and identically distributed
4. sequences are infinitely long
5. Alignments do not contain gaps

Karlin-Altschul Equation:



Dayhoff LOD Score: log odds ratio

$$\text{LOD}(\text{pair of amino acids}) = \log_2 \left(\frac{\text{obs. freq.}}{\text{expected freq.}} \right)$$

$$S_{ij} = \frac{p_{ij}}{p_i p_j}$$

p_i, p_j = frequencies of AAs
 i & j

if obs. freq = exp. freq then $\text{LOD} = 0$. If $\text{LOD} > 0$ then pairing is linked (correlated). If $\text{LOD} < 0$ pairing is more likely to have occurred if pairing is unlinked.